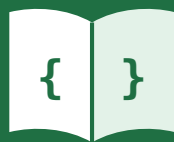

Module 22 — Kubernetes

Exercises

Spring Boot Edu



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How to Work

1. The exercise manifests are in `modules/22-kubernetes/exercise/k8s/`. Find the TODO comments.
2. The tests read your YAML files and check the rules, without a cluster:

```
./mvnw -Pexercises -pl modules/22-kubernetes/exercise -am test
```

3. Then try your manifests in the kind cluster (each exercise has a “Try it” part). They run in their own namespace `bookstore-exercise` and use the lesson’s image (run `modules/22-kubernetes/deploy.sh` once, it loads the image into kind):

```
kubectrl apply -k modules/22-kubernetes/exercise/k8s
kubectrl -n bookstore-exercise rollout status deployment/bookstore
```

4. The exercise is done when all tests are green and the “Try it” parts behave as described.
5. If you get stuck, read the hints first, then the solution in `modules/22-kubernetes/solution/k8s/`.

Exercise 1 — Configuration Changes That Reach the Pods (Easy)

Goal: the shop name comes from a hand-written ConfigMap (`configmap.yaml`). Change it, apply, and the running pods still show the old name.

Tasks (`exercise/k8s/`):

- TODO 1 — replace `configmap.yaml` with a `configMapGenerator` named `bookstore-config` in `kustomization.yaml` (the same key `BOOKSTORE_SHOP_NAME`), and remove `configmap.yaml` from the resources.

Hints:

- Lesson section 3.2. The Deployment keeps `configMapRef: name: bookstore-config`; `Kustomize` replaces it with the generated name.
- Why did the old way not work? Environment variables are read only when the container starts.

Try it: apply, change the shop name in `kustomization.yaml`, apply again, and watch `kubectrl -n bookstore-exercise get pods -w`: new pods start with the new name (`port-forward + /api/shop`).

Acceptance criteria: all 3 tests in `Exercise1Test` pass.

Estimated time: 20 minutes

Exercise 2 — A Broken Version Must Not Hurt (Medium)

Goal: a new version that does not start must not take traffic away, and you must be able to go back.

Tasks (exercise/k8s/deployment.yaml):

- TODO 2a — a rolling update that never removes a ready pod before its replacement is ready, one new pod at a time.
- TODO 2b — a readiness probe on /actuator/health/readiness.
- TODO 2c — keep at least 3 old versions (revisionHistoryLimit).

Hints: lesson sections 3.3 and 3.6 (maxUnavailable, maxSurge).

Try it: set a broken image, look at the pods, then roll back:

```
kubectl -n bookstore-exercise set image deployment/bookstore
↪ bookstore=springbootedu/kubernetes:broken
kubectl -n bookstore-exercise get pods
kubectl -n bookstore-exercise rollout undo deployment/bookstore
```

Write down: how many old pods were ready while the broken pod failed?

Acceptance criteria: all 3 tests in Exercise2Test pass, and the old pods stayed ready during the broken rollout.

Estimated time: 30 minutes

Exercise 3 — Scale Under Load (Medium)

Goal: between 1 and 4 pods, depending on CPU load.

Tasks:

- TODO 3a (deployment.yaml) — request 200m CPU.
- TODO 3b — an autoscaler in hpa.yaml (added to the resources): target the Deployment bookstore, 1 to 4 replicas, 50 % average CPU.

Hints:

- Lesson section 3.6. Without a CPU request, the HPA shows cpu: <unknown> and does nothing.
- The metrics-server comes from scripts/kind-up.sh.

Try it: start the load pod of lesson section 3.6 in the namespace bookstore-exercise and watch `kubectl -n bookstore-exercise get hpa -w`.

Acceptance criteria: both tests in Exercise3Test pass, and the number of replicas goes up under load.

Estimated time: 30 minutes

Extra Challenge (Optional)

Add a `PodDisruptionBudget` (`minAvailable: 1`) and drain the kind node (`kubectl drain bookstore-control-plane --ignore-daemonsets --delete-emptydir-data`). What does the budget change? Undo it with `kubectl uncordon bookstore-control-plane`.